

Experiment :- Study of LAN Topology, Network devices and Address configuration

Objective :-

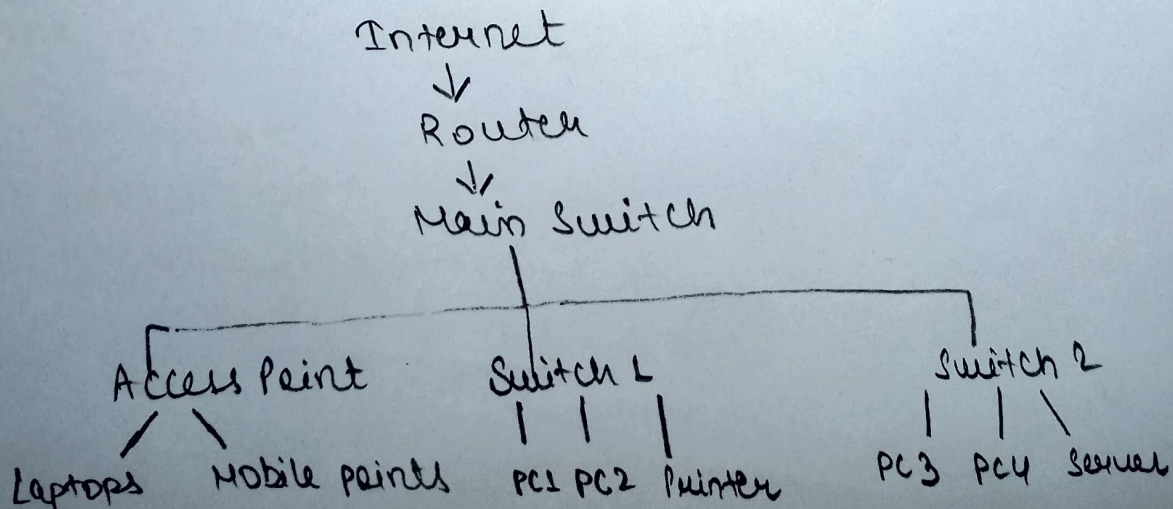
- (1.) To study the LAN topology used in college campus
- (2.) To identify networking device & understand their function.
- (3.) To analyze wired & wireless communication method used in the network.
- (4.) To measure speed of network for wired & wireless connections
- (5.) To obtain MAC, IP Address & subnet mask.

Statement - 1

Draw & identify LAN topology of college campus & clearly label all components such as router, switches & etc.

→ The college campus network observed this study consist of:

- Layer-3 core switches in the server room.
- Managed by POE + Switches
- wireless Access points (APs)
- Ethernet point in classroom & laboratories.



Components Identification: -

(1.) Router

- Connects the organization LAN to the Internet.

(2.) Main Switch

- central connecting device for all network components.

(3.) Access Point (AP)

- Provides wireless network access to laptops/mobile devices

(4.) Switch 1 / Switch 2

- Connect multiple wired devices

(5.) End Devices

- PC1, PC2, P3, PC4 → Computers used by users.
- Printer - shared network printer
- Server - stores data / services
- Laptops/Mobiles phones - wireless devices connected via AP

Network Topology

Hybrid Hierarchical Topology which combines: -

(1.) Tree Topology

(2.) Extended star Topology

(3.) Ring Topology (Server room).

The network hierarchical layer

(1) Core layer

(2) Distribution layer

(3) Access layer.

Core layer

- * Core layer Contains Server Room and Server room have Firewall, Router, 2-L3 + Switches, Patch rack server, inverter etc.

Distribution layer

- * Distribution layer contains 4-switches.
- * The 4 switches are →

- (1) Main Build → R7 (Room no-213, 2nd floor).
- (2) Main Build → R6 (Room no-301, 3rd floor).
- (3) Main Build → R6 (Room no-113, 1st floor).
- (4) Main Build → R2 (Room no:- 201, 2nd floor).

- * This 4-switches contains

- (1) 18 port Gigabit Easy Smart Switch with 16-Port
 - [TL-SG1218MPE]

- (2) SG3428XMP → 24 Port Gigabit X 4 port 10 GbE SFP+, 12+ Managed Switches with 24 Port PoE+

→ Now let's discuss these 2 above components

(1) TP-link TL-SG1218MPE

- * likely used for : CCTV Camera, wifi access point etc.
- * 18 Port Gigabit "Easy Smart" switch.
- * 16 Port support PoE+

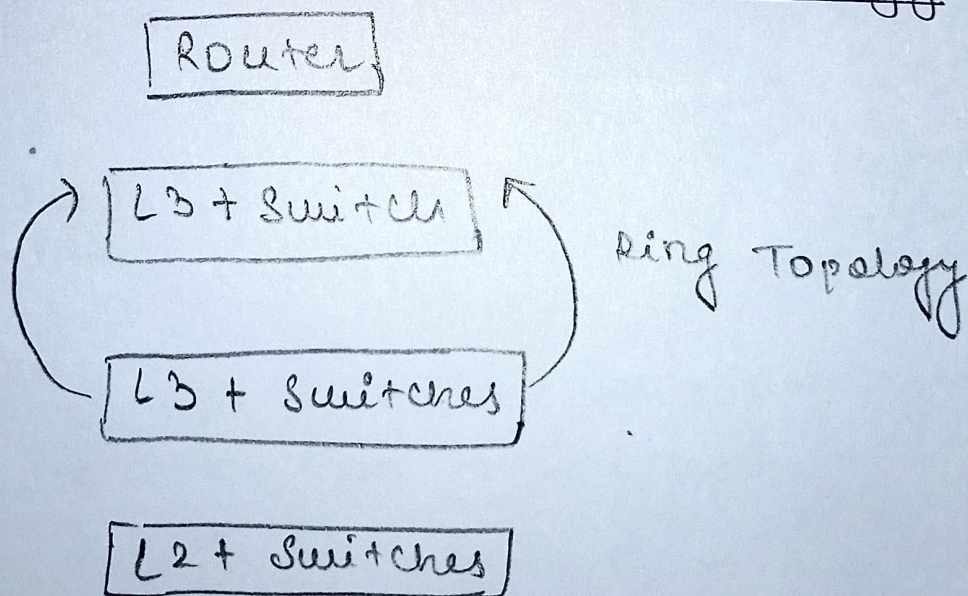
(2) TP-link Omada SG3428XMP

- * Higher-end managed layer 2+ switch
- * 24 PoE+ Port, uplink, 10Gb SFP
- * usually acts as a main distribution for each floor.

Access layer

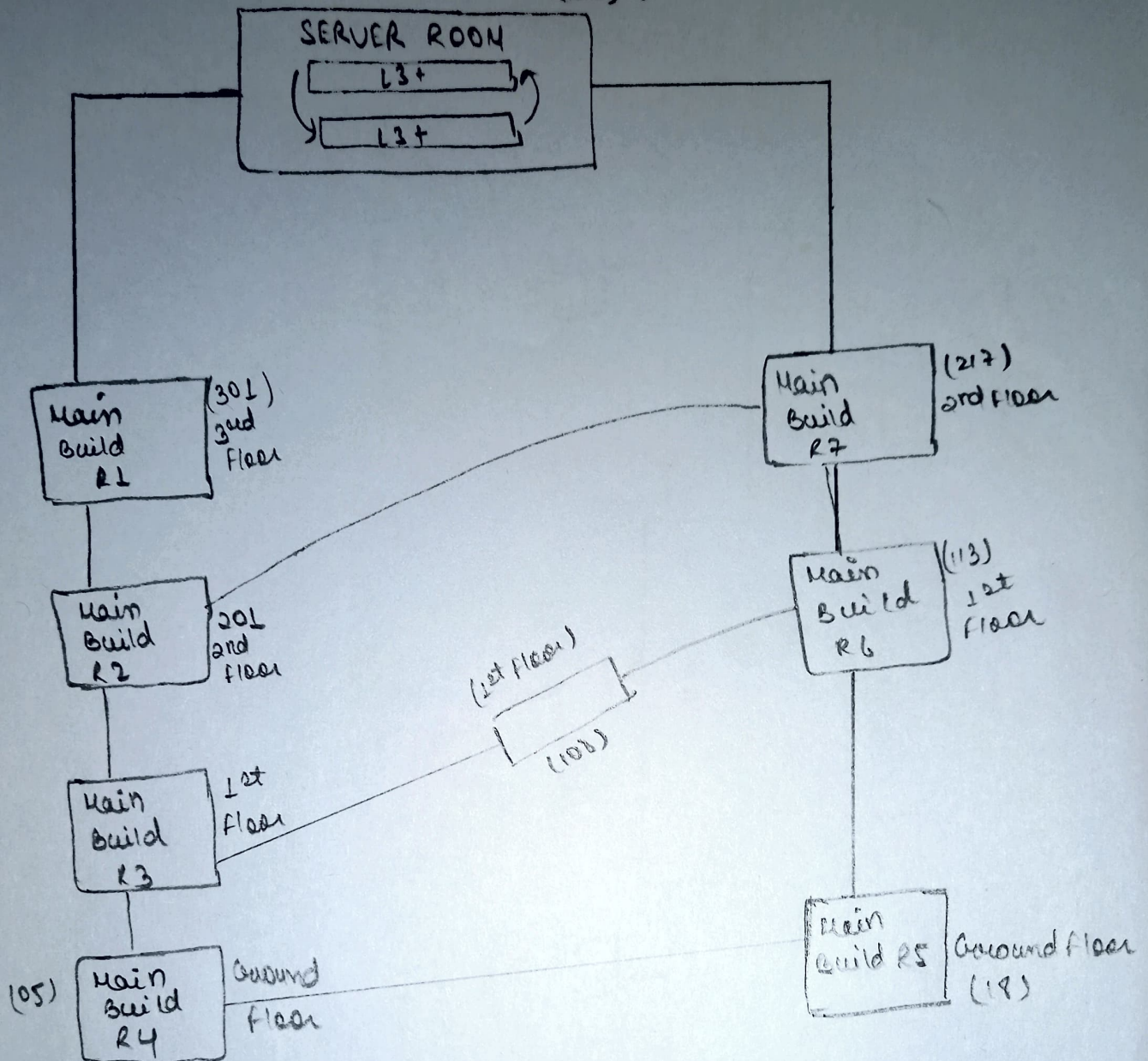
- * Access layer contain 3 switches i.e.
 - (1) Main Build RL - [Room no-301, 3rd floor].
 - (2) Main Build RU - [Room no-05, Ground floor].
 - (3) Main Build RS - [Room no-18, Ground floor].
- * Every switches act as access layer which helps to provide network connectivity to all classroom lab, faculty room, etc.
- * Access layer directly connect:
 - (1) Ethernet port
 - (2) wireless AP
 - (3) Lab system
 - (4) Outdoor wireless AP (ex-018)

SERVER ROOM Architecture / Topology



TOPOLOGY

(210) (2nd floor)

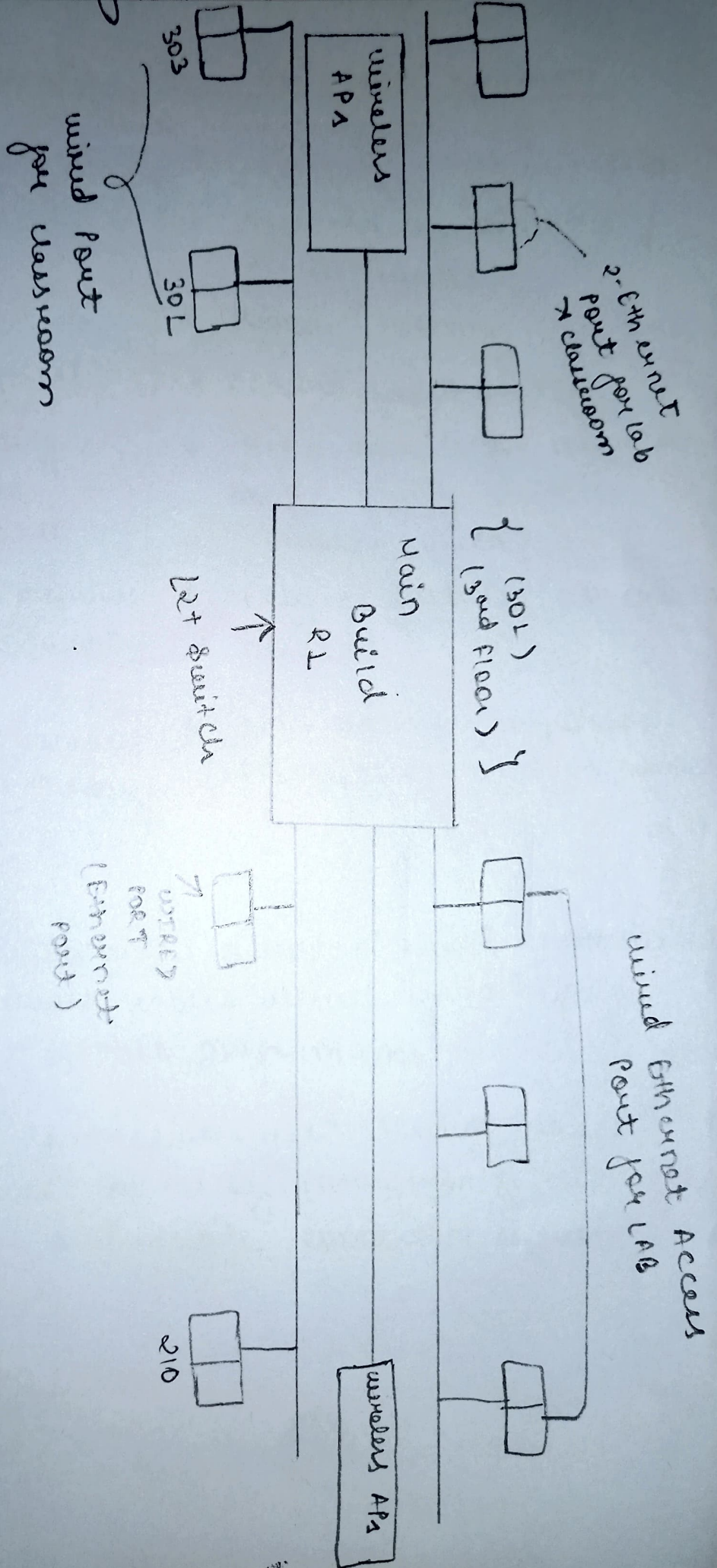


→ R₁, R₇, R₂, R₆ → Distribution layer

→ Server Room → Core layer

→ R₁, R₂, R₃, R₄, R₅, R₆, R₇ → Access layer

Access Layer Switch (3rd floor - 301)





CPU
2% 1.96 GHz



Memory
7.9/15.7 GB (50%)



Disk 0 (C: D: F: E:)
SSD (NVMe)
0%



Wi-Fi
Wi-Fi
S: 0 R: 0 Kbps



GPU 0
Intel(R) UHD Graphics
0%

Wi-Fi

Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter



Send

0 Kbps

Receive

0 Kbps

Adapter name:

Wi-Fi

SSID:

Swastikd

Connection type:

802.11n

IPv4 address:

10.114.172.69

IPv6 address:

2409:40e4:1148:4616:dd9:a710:710:ff1b

Signal strength:



C:\Users\LENOVO>getmac

Physical Address	Transport Name
=====	=====
C0-35-32-4D-BC-15	\Device\Tcpip_{50FFF38C-03E1-4804-97DA-028F99728F22}

C:\Users\LENOVO>

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

```
C:\Users\LENOVO>ipconfig
```

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
```

Wireless LAN adapter Wi-Fi:

```

Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2409:40e4:1148:4616:dd9:a710:710f:ff1b
Temporary IPv6 Address. . . . . : 2409:40e4:1148:4616:e48c:e746:6034:b04e
Link-local IPv6 Address . . . . . : fe80::fec:b58d:9b8c:61c6%10
IPv4 Address. . . . . : 10.114.172.69
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::8994:bfb6:89ae:7ee%10
                             10.114.172.195

```

C:\Users\LENOVO>

Q Difference b/w wired & wireless speed?

→ <u>wired Network</u>	<u>wireless Network</u>
* Faster Communication * Stable connection * low latency * Dedicated physical * limited mobility; devices must be connected by cable * Installation requires cables and network ports * More secure because data travels through cables.	* Slower communication * Affected by distance / Interference * Higher latency * Shared radio medium * High mobility; users can move freely within coverage area. * Easy to install; no cables needed. * less secure; requires encryption and password.

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ wireless connections use radio waves which are affected by interference, distance, obstacles & multiple connected users.

(Q.) Role of subnet mask in network ?

→ A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

* It is important for:-

- ① Network segment
- ② Efficient routing
- ③ communication b/w devices.

OBSERVATION

During the study of the college network infrastructure:-

- ① A centralized server room was used as network backbone.
- ② Two L3 switches were connected in redundant cyclic structure.
- ③ Managed TP-link PoE + switches were used in network cabinets.

Advantages of Existing Network ?

- * Scalable architecture
- * Centralized management
- * Supports POE + devices
- * Easy expansion of floors & classroom
- * Support both wireless & wired
- * Redundant link for failures.

Conclusion

The observed campus LAN follows a Hybrid hierarchical topology, consist of tree, Extended star, Ring. The network uses managed POE switches wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.